STPM 2018 PHYSICS 960/1

Questions and Answers

All calculations use $g = 9.81\ m\ s^{-2}$ and the values of constants printed in the question paper.

Section A [15 marks]

1. In an experiment, a student measures the external diameter, D, and the internal diameter, d, of a plastic tube. The measurement is as below.

$$D = (50.0 \pm 0.2)\ \text{mm}$$
$$d = (48.0 \pm 0.2)\ \text{mm}$$

What is the thickness of the tube?

A (1.0 ± 0.1) mm B (1.0 ± 0.2) mm C (2.0 ± 0.1) mm D (2.0 ± 0.2) mm

Answer: B

$$\text{Thickness} = \frac{D - d}{2} = \frac{50.0 - 48.0}{2} = 1.0\ \text{mm}$$
$$\text{Uncertainty} = \frac{0.2 + 0.2}{2} = 0.2\ \text{mm} \Longrightarrow (1.0 \pm 0.2)\ \text{mm}$$

Explanation: The tube wall is on both sides of the hole, so the thickness is half the difference in diameters. When quantities are subtracted, their absolute uncertainties add; halving then halves the uncertainty.

2. An aeroplane flies horizontally at a constant speed of 100 m s^{-1}. It drops a box of food which lands forward on the ground at 600 m. By neglecting air resistance, the height of the aeroplane from the ground is

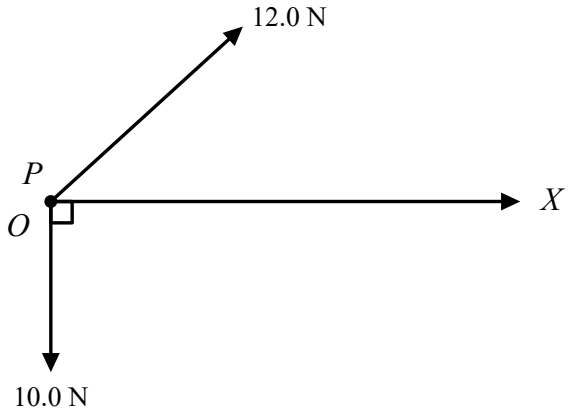
A 177 m B 240 m C 353 m D 500 m

Answer: A

$$\text{Time to fall: } t = \frac{600}{100} = 6.0\ \text{s}$$
$$h = \frac{1}{2}gt^2 = \frac{1}{2}(9.81)(6.0)^2 = 177\ \text{m}$$

Explanation: The box keeps the horizontal speed of the aeroplane, so the horizontal motion gives the time. Vertically it starts from rest and falls freely for that time.

3. Two forces, 12.0 N and 10.0 N, which act on an object, *P*, of mass 2.0 kg are shown in the diagram below.



What is the acceleration of *P* if it moves in the direction of *OX*?

A $1.0\ \text{m s}^{-2}$ B $3.3\ \text{m s}^{-2}$ C $7.8\ \text{m s}^{-2}$ D $11\ \text{m s}^{-2}$

Answer: B

For motion along *OX*, the vertical components must cancel: $12.0 \sin \alpha = 10.0 \Longrightarrow \sin \alpha = 0.833$, $\alpha = 56.4°$

$$F_x = 12.0 \cos 56.4° = 6.63\ \text{N};\ a = \frac{6.63}{2.0} = 3.3\ \text{m s}^{-2}$$

Explanation: The object accelerates only along *OX*, so there is no resultant force perpendicular to *OX*. This fixes the direction of the 12.0 N force; only its component along *OX* accelerates the object.

4. Two satellites orbit the Earth with the same period. Which quantity for both satellites must be of equal values?

A Mass
B Kinetic energy
C Radius of the orbit
D Magnitude of the momentum

Answer: C

$$T^2 = \frac{4\pi^2}{GM} r^3$$

The same period means the same orbital radius.

Explanation: The period depends only on the orbital radius (and the mass of the Earth), not on the mass of the satellite. Mass, kinetic energy and momentum all depend on the satellite's mass, which may differ.

5. An ant of mass 1.2 g climbs up a mirror 25° from the vertical at a speed of 2.3 cm s^{-1} as shown in the diagram below.



What is the output power of the ant?

A 1.1×10^{-4} W B 1.3×10^{-4} W C 2.5×10^{-4} W D 2.7×10^{-4} W

Answer: C

$$\text{Vertical component of velocity} = 2.3 \cos 25° = 2.08 \text{ cm s}^{-1}$$

$$P = mgv \cos 25° = (1.2 \times 10^{-3})(9.81)(2.08 \times 10^{-2}) = 2.5 \times 10^{-4} \text{ W}$$

Explanation: The ant's output power is the rate of gaining potential energy, mg × (rate of rise). The path is 25° from the vertical, so use the vertical component of its velocity. Convert grams and centimetres to SI units.

6. A remote control toy car moves in a vertical circle at a constant speed. Which is true about the linear and angular velocities of the toy car?

	Linear velocity	*Angular velocity*
A	Changes	Constant
B	Constant	Constant
C	Changes	Changes
D	Constant	Changes

Answer: A

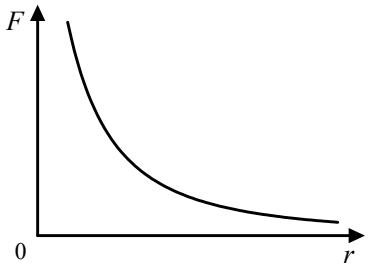
Linear velocity: its direction changes continuously (along the tangent), so it changes.

Angular velocity: constant speed in a circle of fixed radius means $\omega = v/r$ is constant, with a fixed direction (along the axis).

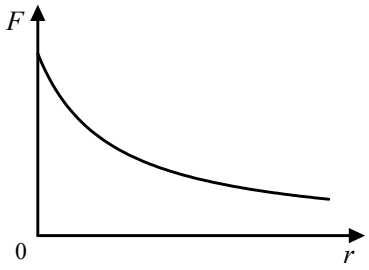
Explanation: Even at constant speed, the linear velocity is not constant because velocity is a vector whose direction changes.

7. Which graph shows the variation of gravitational force, F, between two point masses with separation, r?

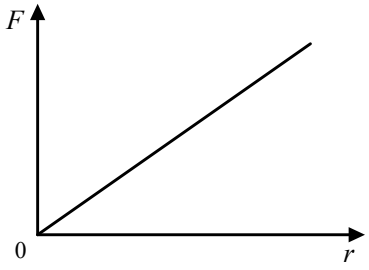
A



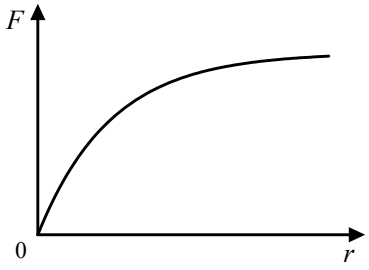
B



C



D



Answer: A

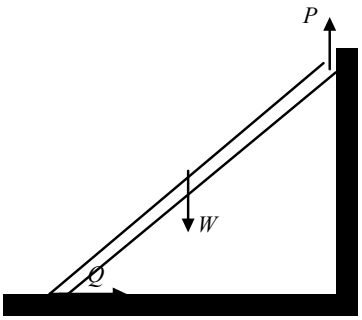
$$F = \frac{Gm_1m_2}{r^2} \Longrightarrow F \propto \frac{1}{r^2}$$

The force becomes very large as $r \rightarrow 0$ and approaches zero as r becomes large.

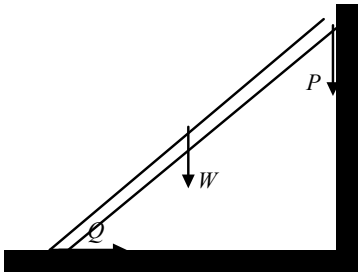
Explanation: An inverse-square curve rises steeply towards the F-axis without crossing it. Graph B starts at a finite value at $r = 0$, which is not correct for point masses.

8. A uniform ladder of weight, W, rests on a rough floor and leans against a rough wall. P and Q are forces which are exerted on the ladder by the wall and floor respectively. Which diagram shows the directions of P, Q and W correctly?

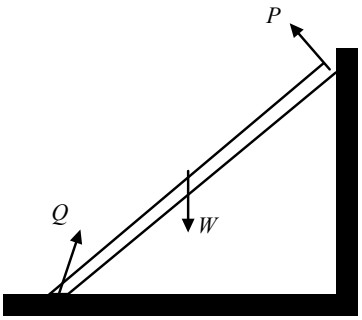
A



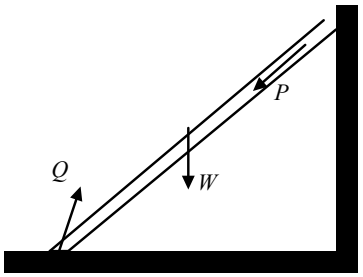
B



C



D



Answer: C

Both surfaces are rough, so each contact force is the resultant of a normal reaction and friction.

P (wall): horizontal normal reaction away from the wall plus upward friction $\Longrightarrow$ P points up and away from the wall.

Q (floor): upward normal reaction plus friction towards the wall $\Longrightarrow$ Q points up and towards the wall. W acts vertically down at the centre.

Explanation: For three non-parallel forces in equilibrium, their lines of action must pass through one point. Only diagram C has P, Q and W in directions that can meet at a common point.

9. The mean translational kinetic energy of oxygen molecules at a temperature of 0 °C is

A 1.88×10^{-21} J B 5.65×10^{-21} J C 9.42×10^{-21} J D 1.32×10^{-20} J

Answer: B

$$\langle E_k \rangle = \frac{3}{2}kT = \frac{3}{2}(1.38 \times 10^{-23})(273) = 5.65 \times 10^{-21} \text{ J}$$

Explanation: The mean **translational** kinetic energy is $\frac{3}{2}kT$ for any gas, even a diatomic one like oxygen. Use 0 °C = 273 K.

10. Two containers, X and Y, are filled with different ideal monatomic gases. The mass of the gas in each container is M and the gases are at the same temperature, T. If the molar masses are M and $2M$ respectively and R is the molar gas constant, which is true about the internal energy of the gas in each container?

	X	Y
A	$\frac{3}{2}RT$	$\frac{3}{4}RT$
B	$\frac{3}{2}RT$	$\frac{5}{2}RT$
C	$\frac{3}{2}RT$	$3RT$
D	$\frac{3}{4}RT$	$\frac{5}{2}RT$

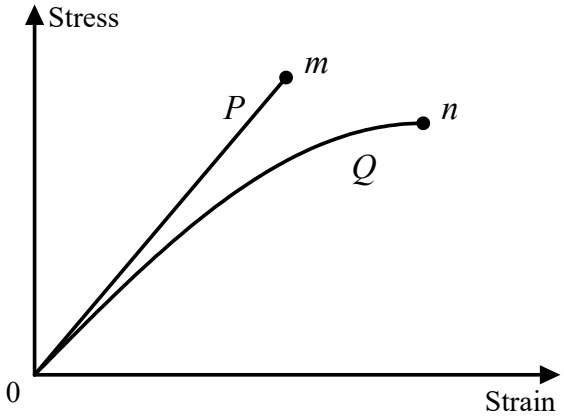
Answer: A

$$X\text{: } n = \frac{M}{M} = 1 \text{ mol} \Longrightarrow U = \frac{3}{2}nRT = \frac{3}{2}RT$$

$$Y\text{: } n = \frac{M}{2M} = \frac{1}{2} \text{ mol} \Longrightarrow U = \frac{3}{4}RT$$

Explanation: The internal energy of a monatomic ideal gas is $\frac{3}{2}nRT$. Equal masses with different molar masses give different numbers of moles.

11. The stress-strain graph of two different materials, *P* and *Q*, are shown below.



Each material fractures at points *m* and *n*. Which statement is **not** true about the properties of *P* and *Q*?

A *Q* has lower tensile strength.
B *P* has greater proportional limit.
C *P* has greater value of Young modulus.
D *Q* has greater energy stored per unit volume for the same strain.

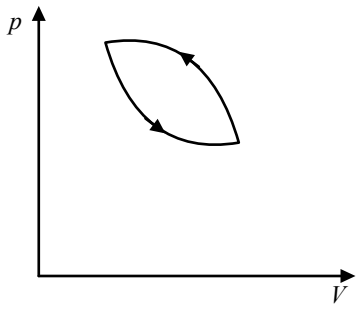
Answer: D

Energy stored per unit volume is the area under the stress–strain graph. For the same strain, *P*'s curve lies above *Q*'s, so *P* stores more energy per unit volume. Statement D is not true.

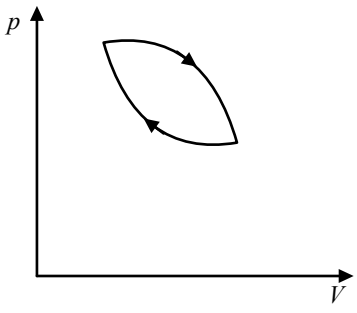
Explanation: *Q* breaks at a lower stress (lower tensile strength), and *P* stays linear further and has the steeper gradient (greater proportional limit and Young's modulus), so A, B and C are true.

12. A fixed mass of gas undergoes the cycle of changes of pressure, *p*, and volume, *V*, as shown in the graph below. If the graphs are drawn with the same scale, which graph shows the net work done by the gas is the largest?

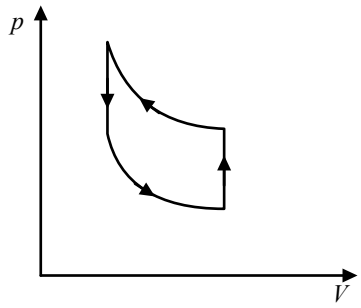
A



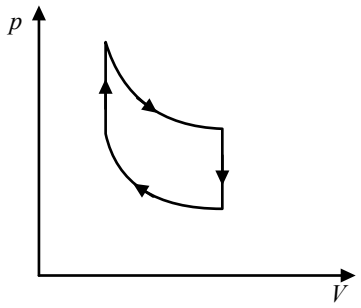
B



C



D



Answer: D

Net work done **by** the gas is positive for a clockwise cycle and equals the enclosed area.
C and D enclose the larger area; D is traversed clockwise, so the net work done by the gas is largest in D.

Explanation: In a clockwise cycle the gas expands at higher pressure than it is compressed, so the work done by the gas exceeds the work done on it. In C the cycle is anticlockwise, so net work is done on the gas.

13. A diatomic gas at initial pressure, p_o, and initial volume, V_o, is compressed adiabatically until its volume is $0.10V_o$. The gas then expands isothermally back to its initial volume, V_o. What is the final pressure of the gas?

A $0.25p_o$ B $0.47p_o$ C $2.5p_o$ D $4.7p_o$

Answer: C

$$\text{Adiabatic: } p_1 = p_o\left(\frac{V_o}{0.10V_o}\right)^{1.4} = 25.1p_o$$

$$\text{Isothermal: } p_2 = 25.1p_o \times \frac{0.10V_o}{V_o} = 2.5p_o$$

Explanation: Use pV^γ = constant with $\gamma = 1.4$ for the diatomic gas in the adiabatic stage, then Boyle's law (pV = constant) for the isothermal stage.

14. Convection is the process of transfer of heat by

A collision of atoms
B diffusion of electrons
C electromagnetic waves
D mass movement of molecules

Answer: D

Convection is the transfer of heat by the mass movement of the fluid molecules themselves.

Explanation: Collisions of atoms and diffusion of electrons describe conduction; electromagnetic waves describe radiation.

15. A black body radiates power of P_o at a temperature of 20.0 °C. What is the power, in terms of P_o, that is radiated when the temperature of the black body is raised to 30.0 °C?

A $1.03P_o$ B $1.14P_o$ C $1.50P_o$ D $5.06P_o$

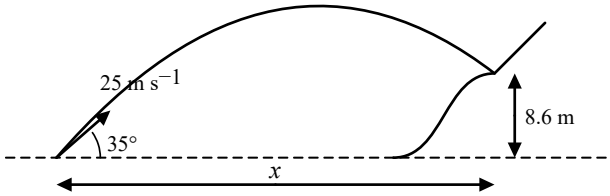
Answer: B

$$\frac{P}{P_o} = \left(\frac{303}{293}\right)^4 = 1.14 \Longrightarrow P = 1.14P_o$$

Explanation: Power radiated $\propto T^4$ with T in kelvin. Using Celsius temperatures, $(30/20)^4 = 5.06$, gives the wrong option D.

Section B [15 marks]

16. A boy kicks a ball and the ball moves with an initial velocity of 25 m s^{-1} at an angle 35° to the horizontal. The ball reaches the maximum height before landing on a hillside as shown in the diagram below.



By neglecting air resistance, determine

(a) the horizontal distance, x, [3]

Answer

$u_x = 25 \cos 35° = 20.48$ m s^{-1}; $u_y = 25 \sin 35° = 14.34$ m s^{-1} — 1

Vertical: $8.6 = 14.34t - \frac{1}{2}(9.81)t^2 \Longrightarrow t = 2.08$ s (on the way down) — 1

$x = u_x t = (20.48)(2.081) = 42.6$ m — 1

3

Explanation: The ball lands 8.6 m above the launch level. The quadratic has two roots (0.42 s and 2.08 s); the ball passes 8.6 m twice, and it lands on the hillside after the maximum height, so take the larger root.

(b) the time of flight of the ball, [1]

Answer

Time of flight $t = 2.08$ s — 1

1

Explanation: This is the larger root found in (a).

(c) the speed of the ball during the impact. [4]

Answer

$v_x = 20.48 \text{ m s}^{-1}$ (unchanged) 1

$v_y = 14.34 - (9.81)(2.081) = -6.07 \text{ m s}^{-1}$ 1

$v = \sqrt{(20.48^2 + 6.07^2)}$ 1

$v = 21.4 \text{ m s}^{-1}$ 1

4

Total for question **8**

Explanation: Combine the horizontal and vertical components at the moment of impact. (Energy gives the same: $v^2 = 25^2 - 2(9.81)(8.6)$.)

17. A cylinder contains 0.25 mol of oxygen gas at a temperature of 27 °C. Assume oxygen gas as an ideal gas.

[Molar mass of oxygen is 32 g mol^{-1}.]

(a) State **two** differences between an ideal gas and a real gas in terms of molecules. [2]

Answer

• Ideal gas: no intermolecular forces except during collisions; real gas: there are intermolecular forces 1

• Ideal gas: volume of the molecules is negligible; real gas: the molecules have a finite volume 1

• Ideal gas: collisions are perfectly elastic; real gas: collisions are not perfectly elastic 1

[Any two – maximum 2 marks]

2

Explanation: Each difference must state both the ideal and the real gas. Note that it is the volume of the **molecules**, not of the gas, that is negligible.

(b) Determine the mean translational kinetic energy of the gas molecules. [2]

Answer

$\langle E_k \rangle = \frac{3}{2}kT = \frac{3}{2}(1.38 \times 10^{-23})(27 + 273)$ 1

$\langle E_k \rangle = 6.21 \times 10^{-21} \text{ J}$ 1

2

Explanation: The mean translational kinetic energy depends only on the absolute temperature.

(c) Calculate the internal energy of the gas. [3]

Answer

Oxygen is diatomic: $f = 5$ 1

$U = \frac{f}{2}nRT = \frac{5}{2}(0.25)(8.31)(300)$ 1

$U = 1.56 \times 10^3$ J 1

3

Total for question **7**

Explanation: The internal energy includes the rotational kinetic energy as well as the translational, so a diatomic gas uses 5 degrees of freedom, not 3. The molar mass is not needed.

Section C [30 marks]

18. (a)

(i) Define *impulse*. [1]

Answer

Impulse is the product of a force and the time interval for which it acts (equal to the change in momentum) 1

1

Explanation: Impulse is a vector, measured in N s (equivalent to kg m s^{-1}).

(ii) Using a suitable Newton law of motion, derive an expression which relates the impulse and velocity. [3]

Answer

Newton's second law: $F = \frac{dp}{dt} = \frac{mv - mu}{t}$ 1

$Ft = mv - mu$ 1

Impulse = $mv - mu$, the change in momentum 1

3

Explanation: Start from Newton's second law in its momentum form (force = rate of change of momentum), then multiply by the time interval.

(b) A piece of plasticine and a ball bearing of the same mass are dropped vertically downward from the same height onto a hard metal surface.

(i) Explain, whether the plasticine or the ball bearing which would exert a larger average force on the surface. [3]

Answer

The ball bearing exerts the larger average force	1
It rebounds, so its change in momentum is about twice that of the plasticine, which stops	1
Its impact time is also shorter (hard material), and $F = \frac{\Delta p}{\Delta t}$, so the force is larger	1
	3

Explanation: Both hit the surface with the same speed. The plasticine deforms and stops (momentum change mv), while the ball bearing bounces back (momentum change up to $2mv$).

(ii) Calculate the average force for the plasticine and the ball bearing if the mass is 10.0 g, the height is 2.0 m and the duration of impact is 0.02 s. [5]

Answer

Speed before impact: $v = \sqrt{(2gh)} = \sqrt{(2(9.81)(2.0))} = 6.26$ m s^{-1}	1
Plasticine (stops): $\Delta p = (0.0100)(6.26) = 0.0626$ kg m s^{-1}	1
$F = \frac{0.0626}{0.02} = 3.13$ N	1
Ball bearing (rebounds with the same speed): $\Delta p = 2(0.0100)(6.26) = 0.125$ kg m s^{-1}	1
$F = \frac{0.125}{0.02} = 6.26$ N	1
	5

Explanation: Assume the ball bearing rebounds elastically with the same speed, so its velocity changes from −6.26 to +6.26 m s^{-1}. The same impact time is given for both.

(c) A bullet of mass 125 g is fired horizontally with a velocity of 315 m s^{-1} into a block of wood of mass 1.50 kg which initially resting on a smooth surface. Calculate the speed of the block after the collision if the bullet is embedded in the block. [3]

Answer

Conservation of momentum: $m_b u = (m_b + M)v$	1
$(0.125)(315) = (0.125 + 1.50)v$	1
$v = 24.2$ m s^{-1}	1
	3
Total for question	**15**

Explanation: The bullet is embedded, so the bullet and block move together afterwards (perfectly inelastic collision). Use 125 g = 0.125 kg.

19. (a) Derive a relationship between linear velocity, v, and angular velocity, ω, for an object moving in a circle of radius, r. [3]

Answer

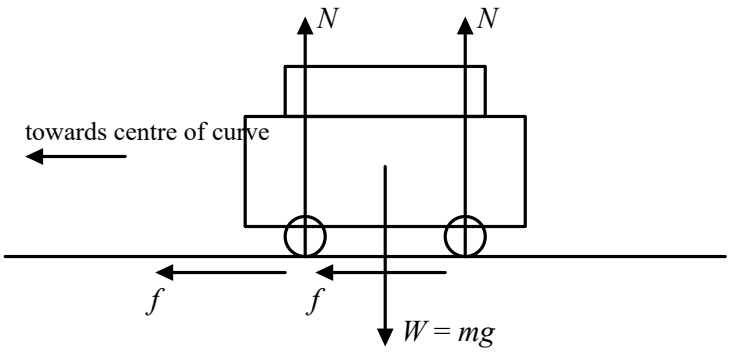
Arc length $s = r\theta$	1
$v = \frac{ds}{dt} = r\frac{d\theta}{dt}$ (r constant)	1
$\omega = \frac{d\theta}{dt} \Longrightarrow v = r\omega$	1
	3

Explanation: The angle must be in radians for $s = r\theta$ to hold.

(b) A car moves with constant speed along a horizontal curve of radius 60.0 m. The coefficient of static friction between the road and the tyres is 0.72.

(i) Draw a labelled diagram to show the forces acting on the car as it negotiates the curve. [2]

Answer



Weight mg downward and normal reaction N upward	1
Friction f on the tyres, horizontal, towards the centre of the curve	1
	2

Explanation: On a flat (unbanked) road the only horizontal force on the car is static friction, so it must point towards the centre of the curve to provide the centripetal force.

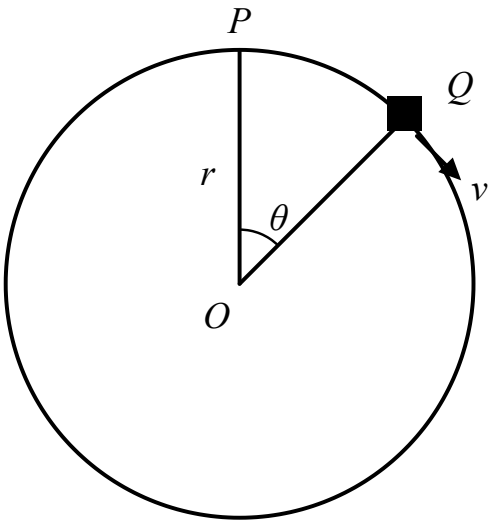
(ii) Determine the maximum speed of the car without skidding. [4]

Answer

Friction provides the centripetal force: $f = \frac{mv^2}{r}$	1
Maximum speed when $f = \mu_s N = \mu_s mg$	1
$v = \sqrt{(\mu_s gr)} = \sqrt{((0.72)(9.81)(60.0))}$	1
$v_{max} = 20.6 \text{ m s}^{-1}$	1
	4

Explanation: Above this speed the friction needed is more than the maximum static friction, so the car skids outwards. The mass of the car cancels.

(c) A mass which initially at rest slides down from the highest point, P, of a smooth sphere of radius, r, as shown in the diagram below.



When the mass reaches at Q, the angular displacement is θ.

(i) Deduce the speed, v, of the mass at Q in terms of r, g and θ. [3]

Answer

Height fallen from P to Q: $h = r - r\cos\theta$	1
$\frac{1}{2}mv^2 = mgr(1 - \cos\theta)$	1
$v = \sqrt{(2gr(1 - \cos\theta))}$	1
	3

Explanation: The sphere is smooth, so mechanical energy is conserved. The vertical drop is measured from the top of the sphere.

(ii) Calculate the angular displacement, θ, when the mass flies off from the surface of the sphere. [3]

Answer

Radial forces at Q: $mg\cos\theta - N = \frac{mv^2}{r}$; it flies off when $N = 0$ — 1

$g\cos\theta = \frac{v^2}{r} = 2g(1-\cos\theta) \Longrightarrow \cos\theta = \frac{2}{3}$ — 1

$\theta = 48.2°$ — 1

3

Total for question **15**

Explanation: The mass leaves the surface when the normal reaction becomes zero: only the component of the weight towards the centre then provides the centripetal force. The answer does not depend on r, g or the mass.

20. (a) Explain the mechanism of heat conduction through a metal. [3]

Answer

At the hot end the free electrons gain kinetic energy and the atoms vibrate with larger amplitude — 1

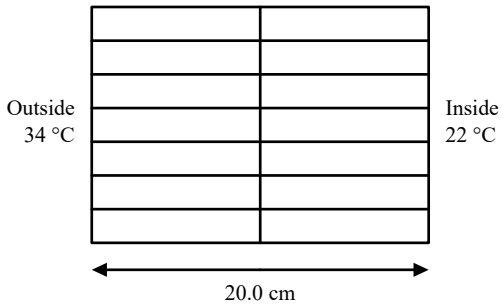
The free electrons diffuse quickly to the cooler region and transfer energy by colliding with atoms there — 1

Energy is also passed by vibrating atoms to neighbouring atoms (lattice vibrations); the electron mechanism is dominant in metals — 1

3

Explanation: Free electrons do not collide with each other to transfer the energy; they transfer it to the atoms (lattice) of the cooler region.

(b) A part of the brick wall of a house is shown in the diagram below.



The wall consists of two layers of bricks in which each layer is of 10.0 cm wide. The outside temperature is 34 °C and the room temperature is maintained at 22 °C. The thermal conductivity

of brick is 0.80 W m^{-1} K^{-1}. Calculate the rate of heat conduction per unit area through the wall. [2]

Answer

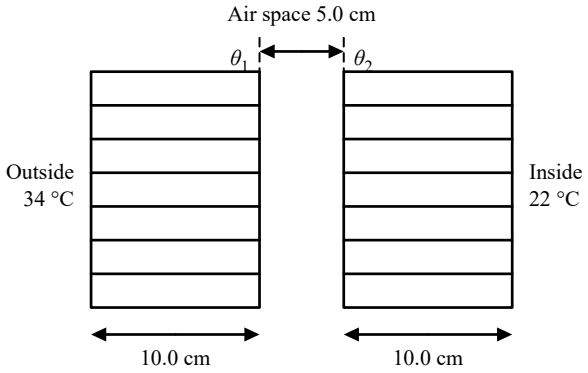
$\frac{dQ/dt}{A} = k\frac{\Delta\theta}{x} = (0.80)\frac{34-22}{0.200}$ 1

$= 48$ W m^{-2} 1

2

Explanation: The two layers of brick together are 20.0 cm = 0.200 m thick.

(c) The brick wall in (b) is separated by an air space with the thickness of 5.0 cm as shown in the diagram below.



The thermal conductivity of air is 0.023 W m^{-1} K^{-1}.

(i) Determine the percentage change in the rate of heat conduction per unit area compared to that of (b). [3]

Answer

$\frac{dQ/dt}{A} = \frac{\Delta\theta}{\Sigma(x/k)} = \frac{12}{\frac{0.200}{0.80}+\frac{0.050}{0.023}} = \frac{12}{2.424}$ 1

$= 4.95$ W m^{-2} 1

Percentage change $= \frac{48-4.95}{48} \times 100\% = 89.7\%$ decrease 1

3

Explanation: The bricks and air are in series, so add their thermal resistances x/k. The 5.0 cm of air has almost ten times the resistance of 20 cm of brick.

(ii) Determine the temperatures for θ_1 and θ_2. [2]

Answer

$\theta_1 = 34 - (4.95)\frac{0.100}{0.80} = 33.4\ °C$ 1

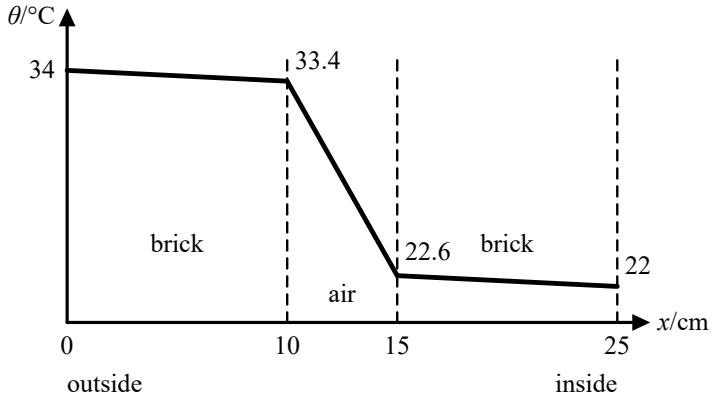
$\theta_2 = 22 + (4.95)\frac{0.100}{0.80} = 22.6\ °C$ 1

2

Explanation: The same heat flow per unit area passes through each layer, so the temperature drop across one brick layer is $(q)(x/k) = 0.62\ °C$.

(iii) Sketch a labelled graph to show how temperature varies across the wall from outside to inside of the house. [3]

Answer



Straight-line, small temperature drops across each brick layer 1

Steep straight-line drop across the air space 1

Temperatures labelled: 34, 33.4, 22.6 and 22 °C (outside to inside) 1

3

Explanation: The temperature gradient in each layer is inversely proportional to its thermal conductivity, so the gradient is steepest in the air, the poorest conductor.

(d) Why polystyrene sheet is a better alternative compared to air although polystyrene has the same thermal conductivity as air? [2]

Answer

Polystyrene traps air in many small pockets (cells) 1

so convection currents cannot form; heat crosses only by conduction (in an open air space convection would increase the heat flow) 1

2

Total for question **15**

Explanation: Still air is an excellent insulator, but in a wide air space the air can circulate. Polystyrene keeps the air still, and it is also a solid that fills and supports the cavity.